

Solving Nonlinear Inequalities (Polynomials) Course

Review on Linear Inequalities

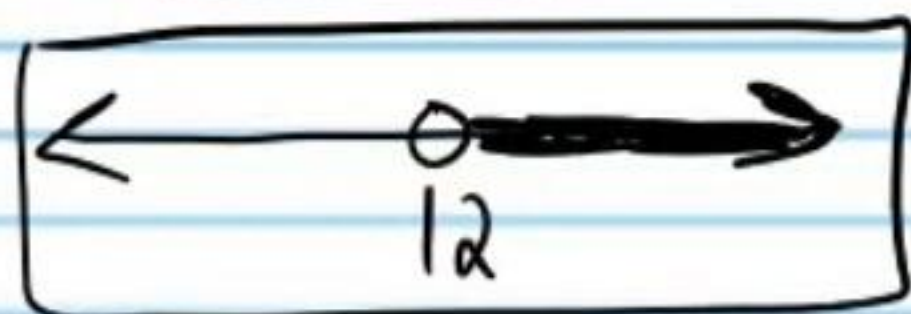
Solve. Write the algebraic solution and the graphical solution.

① $4 + 8 < \frac{x}{2} + 6$

$$12 < \frac{x}{2} + 6$$

$$6 < \frac{x}{2}$$

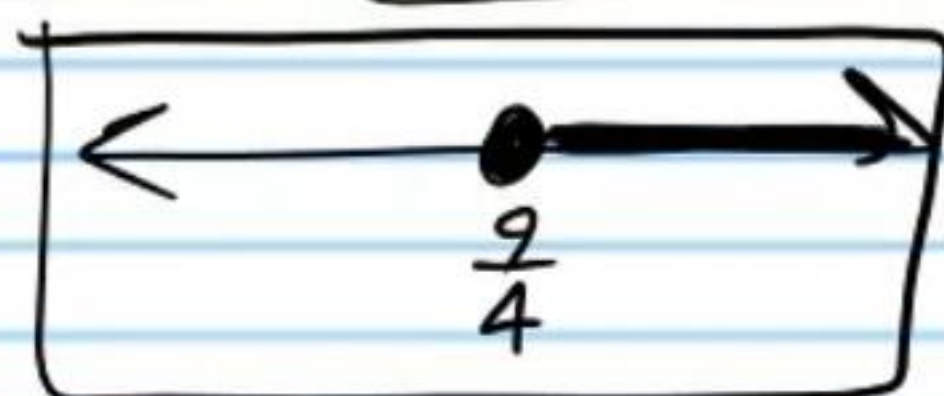
$$12 < x$$



② $10 - 4x \leq 1$

$$-4x \leq -9$$

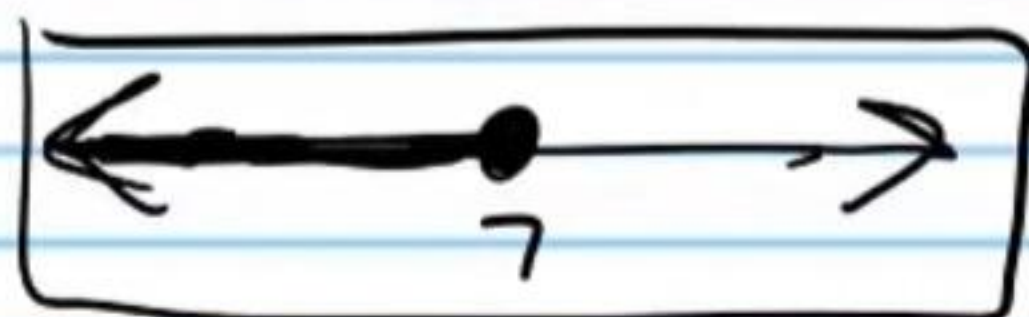
$$x \geq \frac{9}{4}$$



③ $7x - 2x \leq 4x + 7$

$$5x \leq 4x + 7$$

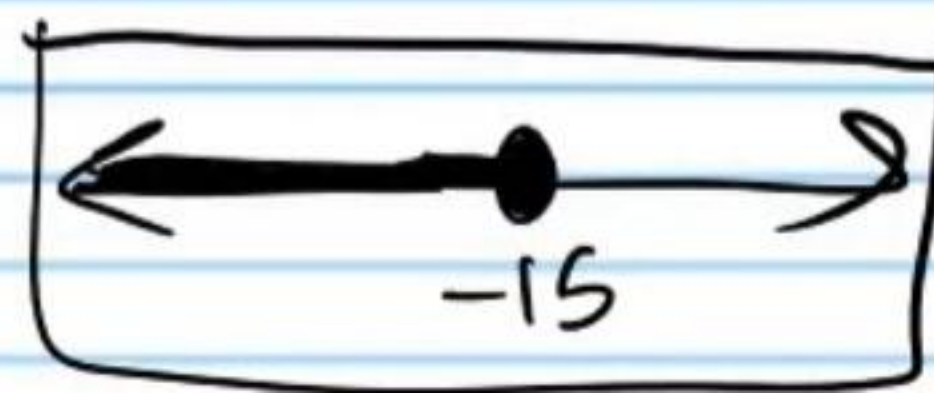
$$x \leq 7$$



④ $-\frac{x}{5} + 1 \geq 4$

$$-\frac{x}{5} \geq 3$$

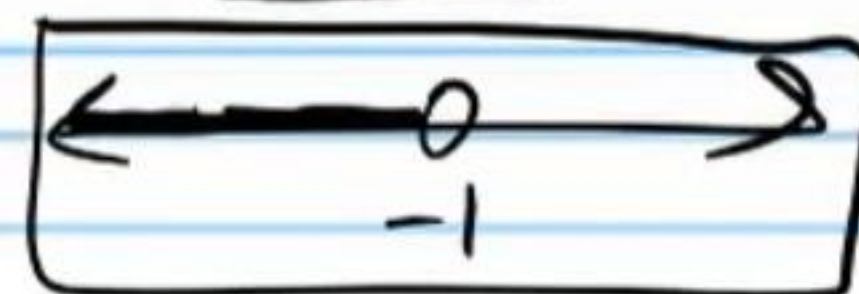
$$x \leq -15$$



⑤ $4 + x < -2x + 1$

$$3x < -3$$

$$x < -1$$

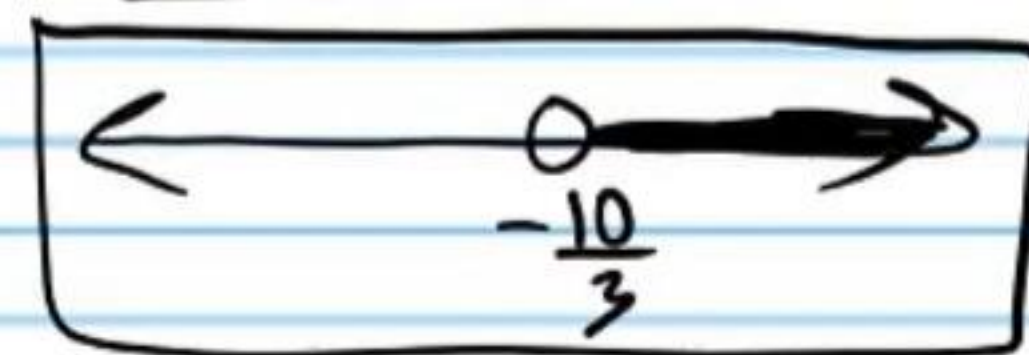


⑥ $8 > 4 - 3x - 6$

$$8 > -2 - 3x$$

$$10 > -3x$$

$$-\frac{10}{3} < x$$



Types of Factoring

Greatest Common Factor $\Rightarrow$ Ex: $10x^3 + 5x - 20x^2 = 5x(2x^2 + 1 - 4x)$

Quadratic Expressions $\Rightarrow$ Ex: $x^2 - 5x - 6 = (x - 6)(x + 1)$

Difference of Two Squares $\Rightarrow$ Ex: $x^2 - 49 = (x + 7)(x - 7)$

$ax^4 + bx^2 + c$ Form $\Rightarrow$ Ex: $x^4 + 5x^2 + 6 = (x^2 + 3)(x^2 + 2)$

Grouping $\Rightarrow$ Ex: $x^3 - 2x^2 + 2x - 4 = x^2(x - 2) + 2(x - 2) = (x - 2)(x^2 + 2)$

Rational Zeros Theorem $\Rightarrow$ Ex: $x^3 - 2x^2 - 5x + 6$

$P: 6 \Rightarrow 1, 2, 3, 6$

$\frac{P}{q}: \pm 1, \pm 2, \pm 3, \pm 6$

$q: 1 \Rightarrow 1$

$$\begin{array}{r|rrrr} 1 & 1 & -2 & -5 & 6 \\ & & 1 & -1 & -6 \\ \hline & 1 & -1 & -6 & 0 \end{array}$$

$$x^3 - 2x^2 - 5x + 6 = (x - 1)(x^2 - x - 6)$$

Sum or Difference of Two Cubes $\Rightarrow$ Ex: $x^3 - 64 = (x - 4)(x^2 + 4x + 16)$

Nonlinear Inequalities

Steps for Solving a Nonlinear Inequality

- I) Move all of the terms to one side of the inequality.
- II) If there is a rational expression (i.e. a fraction with variables in the denominator), combine all of the terms into one fraction.
- III) Find the x-values where the expression is either zero or undefined, and mark these values on a number line.
- IV) Determine the sign of each interval, and use the number line to write the graphical solution of the inequality. Other forms of the solution can be written also. The form most commonly used is interval notation.

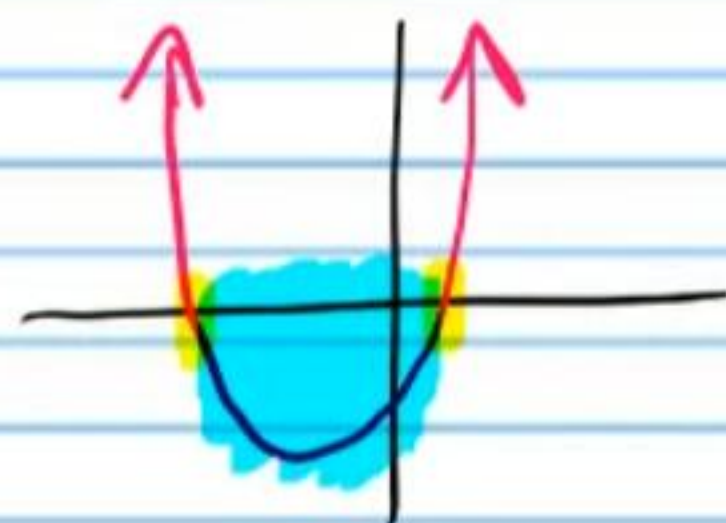
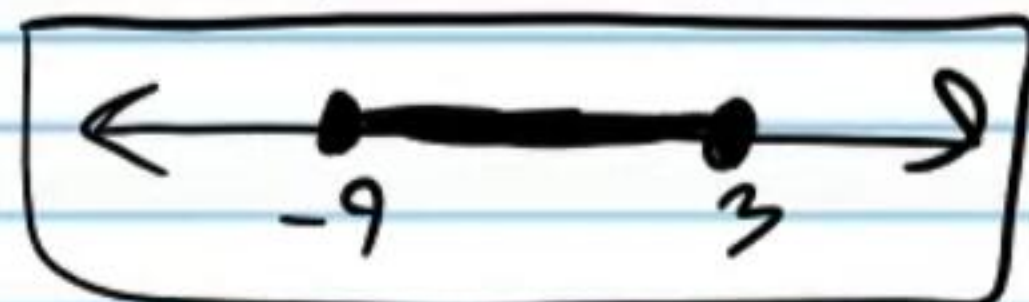
Solve. Write the solutions graphically and in interval notation.

⑦ $27 - 6x \geq x^2$

$y = x^2 + 6x - 27$

$0 \geq x^2 + 6x - 27$

$0 \geq (x+9)(x-3)$

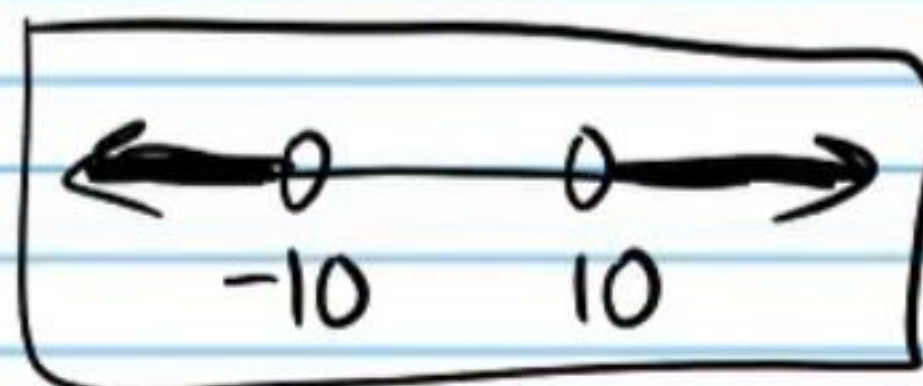
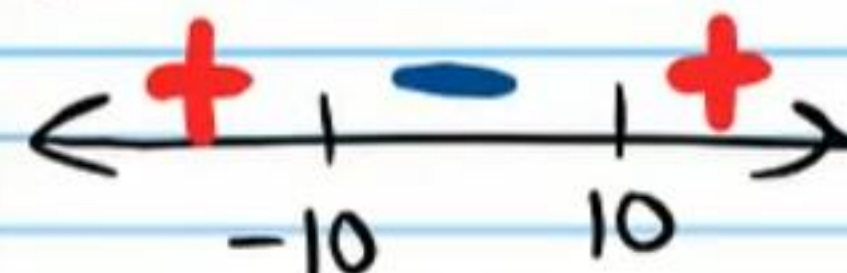


$[-9, 3]$

⑧ $x^2 > 100$

$x^2 - 100 > 0$

$(x+10)(x-10) > 0$



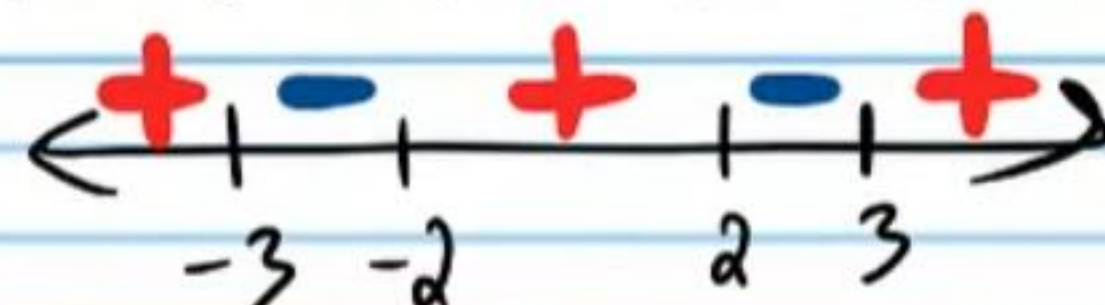
$(-\infty, -10) \cup (10, \infty)$

⑩ $x^4 + 36 \leq 13x^2$

$x^4 - 13x^2 + 36 \leq 0$

$(x^2 - 9)(x^2 - 4) \leq 0$

$(x+3)(x-3)(x+2)(x-2) \leq 0$



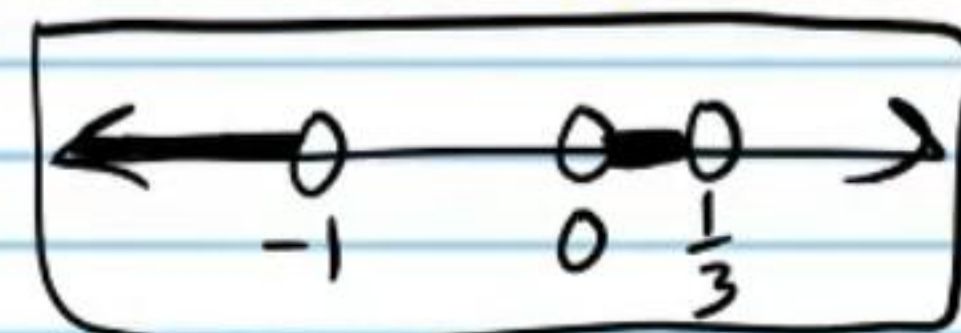
$[-3, -2] \cup [2, 3]$

⑨ $4x^2 < 2x - 6x^3$

$6x^3 + 4x^2 - 2x < 0$

$2x(3x^2 + 2x - 1) < 0$

$2x(3x-1)(x+1) < 0$

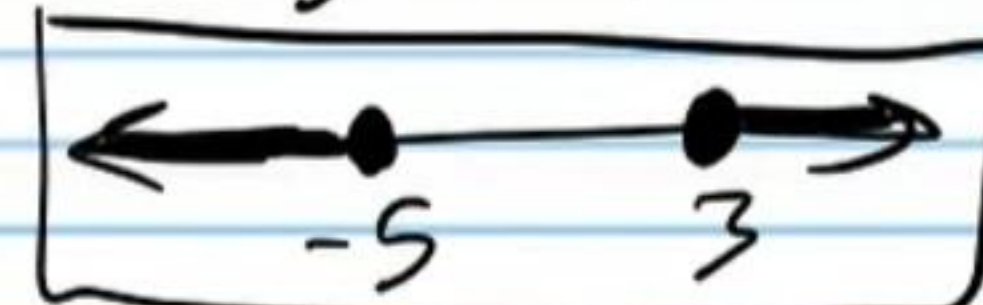
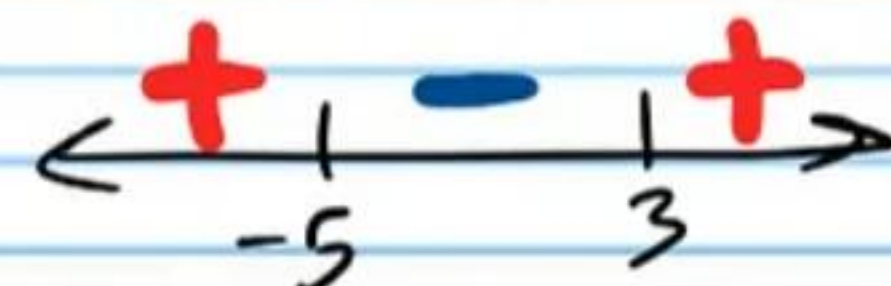


$(-\infty, -1) \cup (0, \frac{1}{3})$

⑪ $x^2 + 2x \geq 15$

$x^2 + 2x - 15 \geq 0$

$(x+5)(x-3) \geq 0$

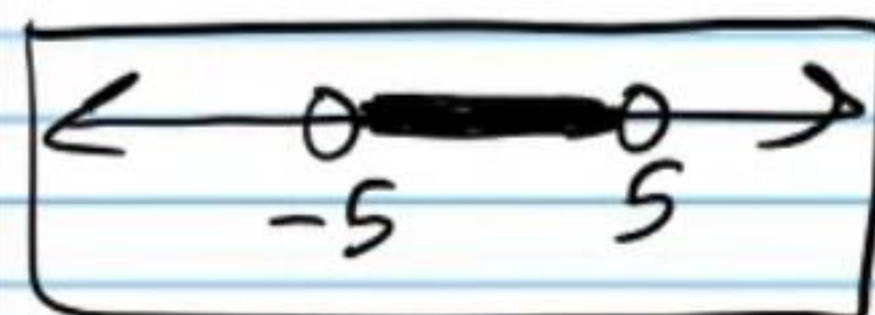
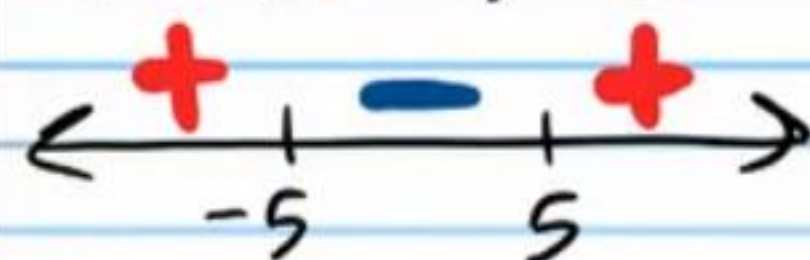


$(-\infty, -5] \cup [3, \infty)$

⑫ $x^2 < 25$

$$x^2 - 25 < 0$$

$$(x+5)(x-5) < 0$$

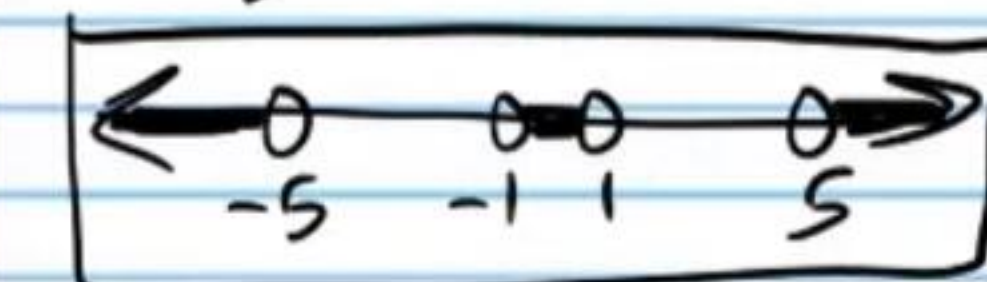
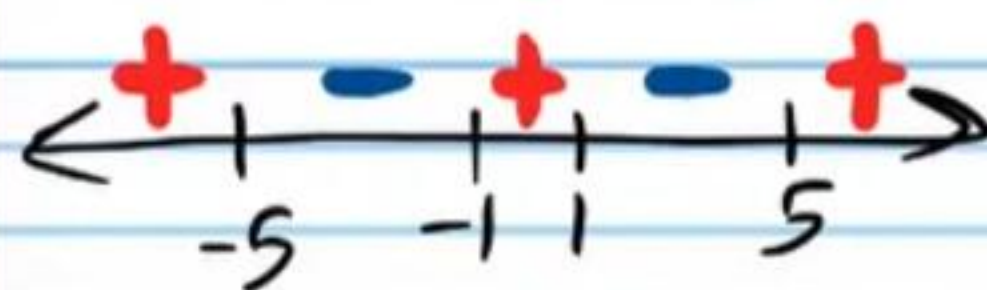


$$(-5, 5)$$

⑭ $x^4 - 26x^2 + 25 > 0$

$$(x^2 - 1)(x^2 - 25) > 0$$

$$(x+1)(x-1)(x+5)(x-5) > 0$$



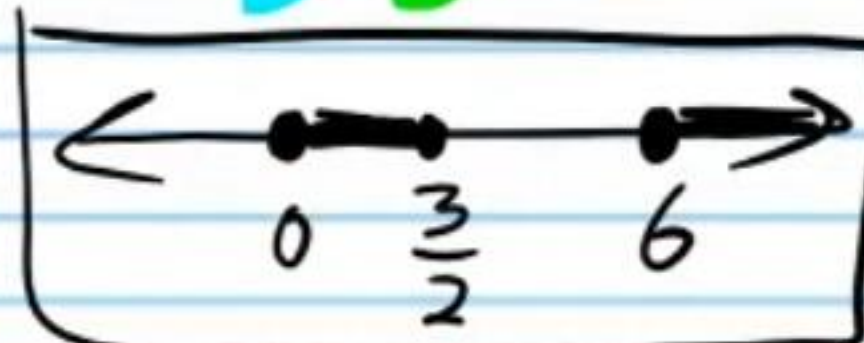
$$(-\infty, -5) \cup (-1, 1) \cup (5, \infty)$$

⑬ $-2x^3 - 18x \leq -15x^2$

$$0 \leq 2x^3 - 15x^2 + 18x$$

$$0 \leq x(2x^2 - 15x + 18)$$

$$0 \leq x(2x-3)(x-6)$$

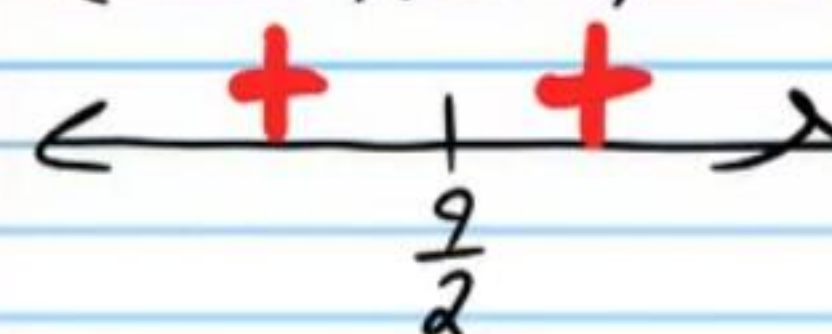


$$\left[0, \frac{3}{2}\right] \cup [6, \infty)$$

⑮ $4x^2 + 81 < 36x$

$$4x^2 - 36x + 81 < 0$$

$$(2x-9)(2x-9) < 0$$



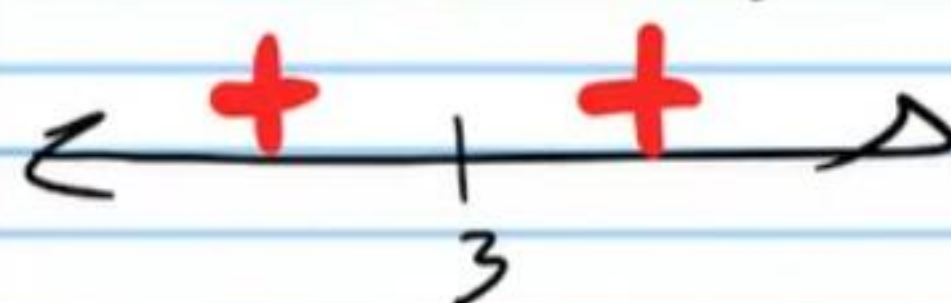
$$\text{No solution}$$

⑰ $6x - 9 \leq x^2$

$$0 \leq x^2 - 6x + 9$$

$$0 \leq (x-3)(x-3)$$

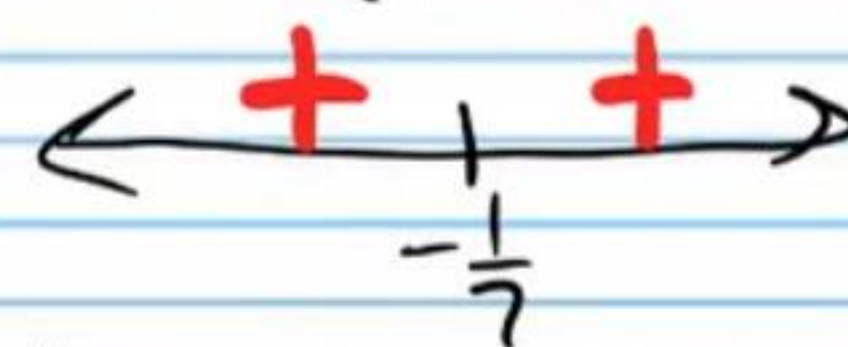
$$0 \leq (x-3)^2$$



$$(-\infty, \infty)$$

⑯ $0 \geq 49x^2 + 14x + 1$

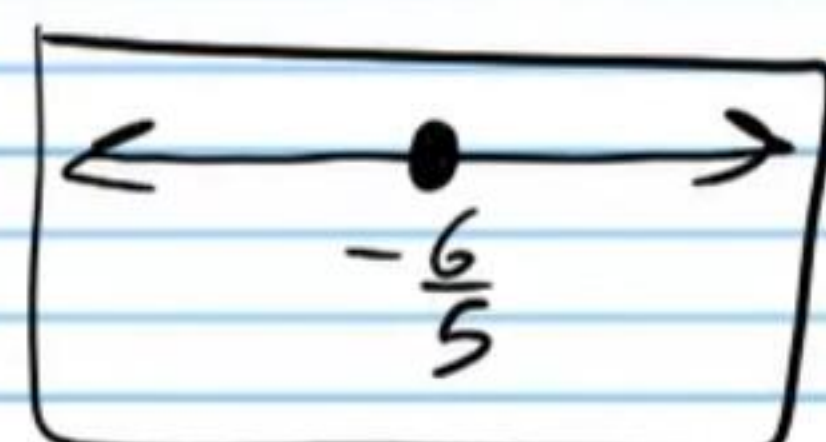
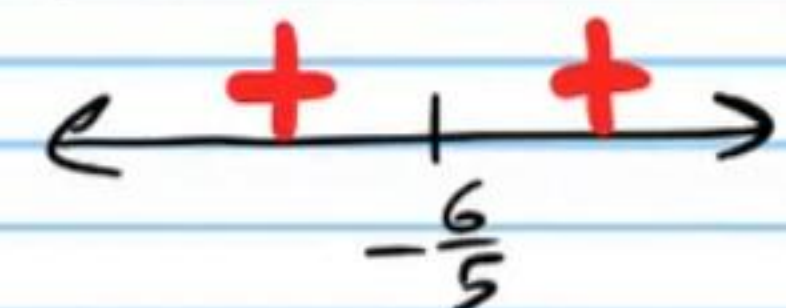
$$0 \geq (7x+1)(7x+1)$$



$$\left[-\frac{1}{7}, -\frac{1}{7}\right]$$

⑱ $0 \geq 25x^2 + 60x + 36$

$$0 \geq (5x+6)(5x+6)$$



$$\left[-\frac{6}{5}, -\frac{6}{5}\right]$$

Using the Rational Zeros Theorem

• (19) $x^3 + x^2 < 8x + 12$

$$x^3 + x^2 - 8x - 12 < 0$$

$p: 12 \Rightarrow 1, 2, 3, 4, 6, 12$

$q: 1 \Rightarrow 1$

$\frac{p}{q}: \pm 1, \pm 2, \pm 3 \rightsquigarrow$

$1 \Rightarrow \text{X} \quad -1 \Rightarrow \text{X}$

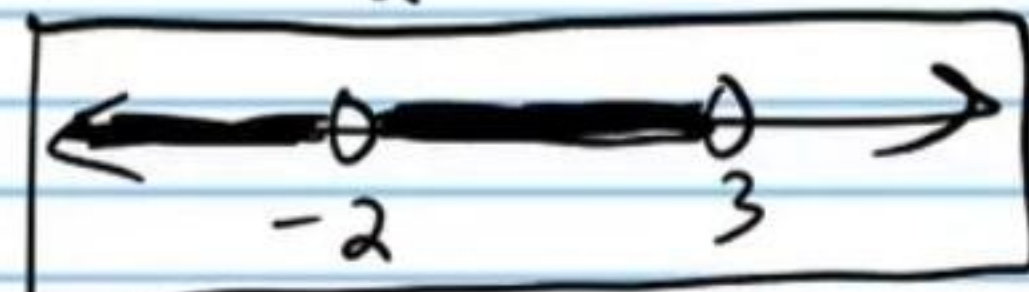
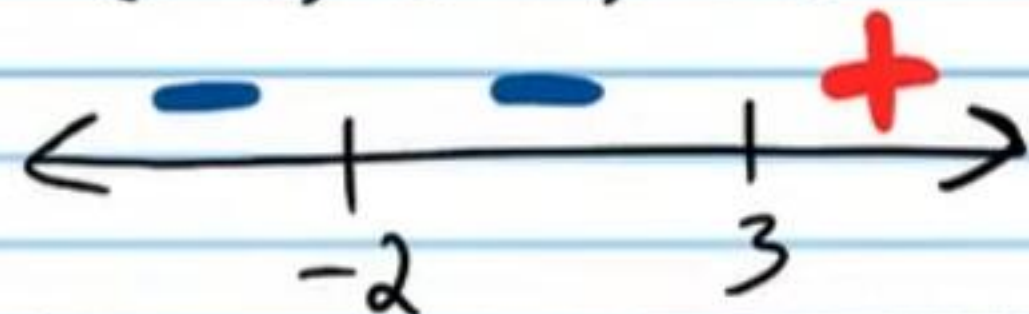
$2 \Rightarrow \text{X} \quad -2 \Rightarrow \checkmark$

$$\begin{array}{r|rrrr} -2 & 1 & 1 & -8 & -12 \\ & & -2 & 2 & 12 \\ \hline & 1 & -1 & -6 & 0 \end{array}$$

$$(x+2)(x^2 - x - 6) < 0$$

$$(x+2)(x-3)(x+2) < 0$$

$$(x+2)^2(x-3) < 0$$



$$(-\infty, -2) \cup (-2, 3)$$

• (20) $x + x^3 + 4x^2 \geq 6$

$$x^3 + 4x^2 + x - 6 \geq 0$$

$p: 6 \Rightarrow 1, 2, 3, 6$

$q: 1 \Rightarrow 1$

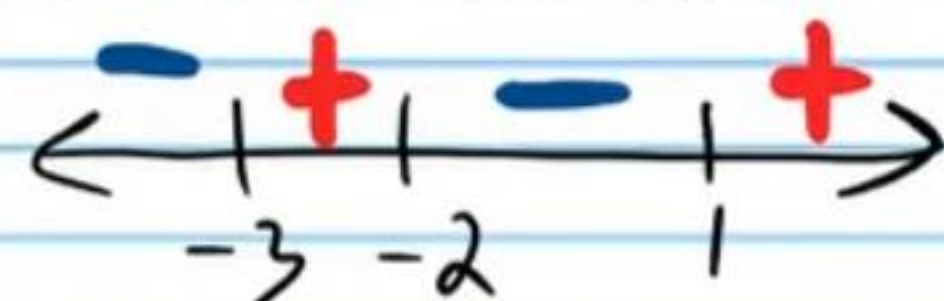
$\frac{p}{q}: \pm 1, \pm 2, \pm 3, \pm 6$

$1 \Rightarrow \checkmark$

$$\begin{array}{r|rrrr} 1 & 1 & 4 & 1 & -6 \\ & & 1 & 5 & 6 \\ \hline & 1 & 5 & 6 & 0 \end{array}$$

$$(x-1)(x^2 + 5x + 6) \geq 0$$

$$(x-1)(x+3)(x+2) \geq 0$$



$$[-3, -2] \cup [1, \infty)$$

Factoring by Grouping

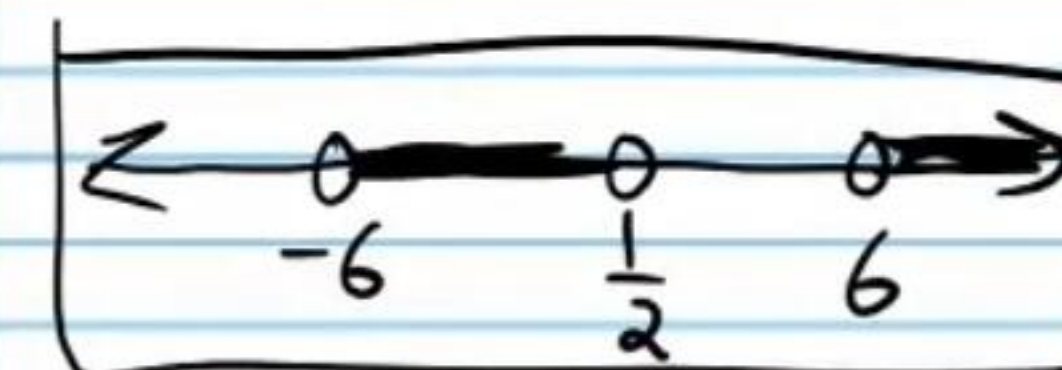
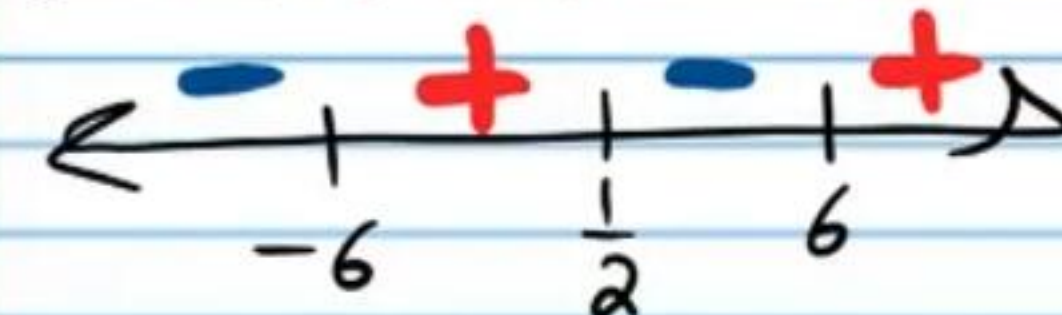
• (21) $2x^3 + 36 > x^2 + 72x$

$$2x^3 - x^2 - 72x + 36 > 0$$

$$x^2(2x-1) - 36(2x-1) > 0$$

$$(2x-1)(x^2 - 36) > 0$$

$$(2x-1)(x+6)(x-6) > 0$$



$$(-6, \frac{1}{2}) \cup (6, \infty)$$

• (22) $x^3 + 1 \leq x^2 + x$

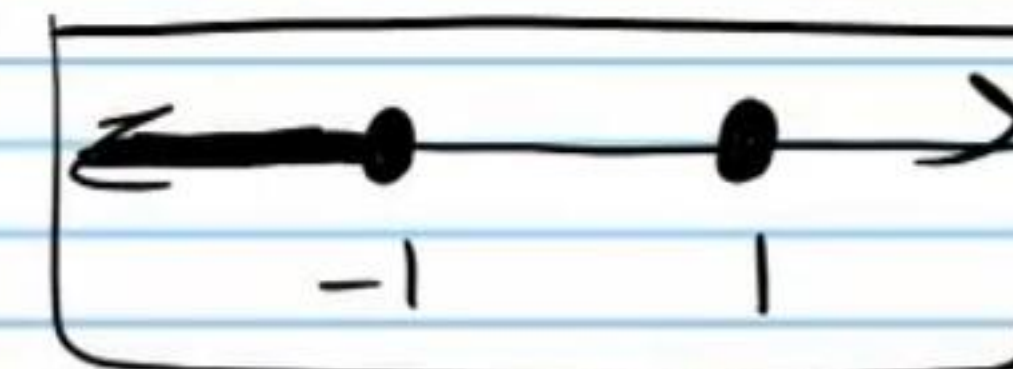
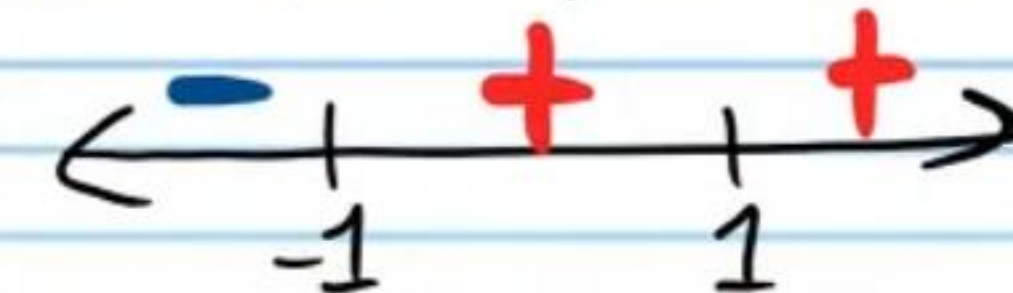
$$x^3 - x^2 - x + 1 \leq 0$$

$$x^2(x-1) - (x-1) \leq 0$$

$$(x-1)(x^2 - 1) \leq 0$$

$$(x-1)(x+1)(x-1) \leq 0$$

$$(x-1)^2(x+1) \leq 0$$



$$(-\infty, -1] \cup [1, 1]$$

Irrational Zeros

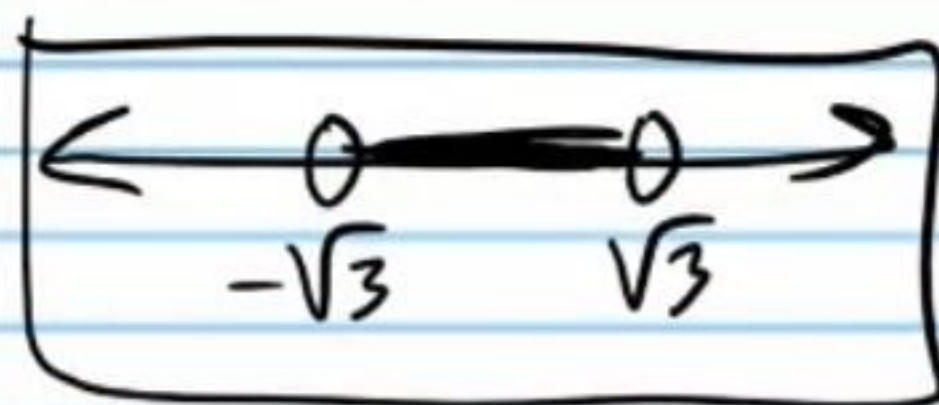
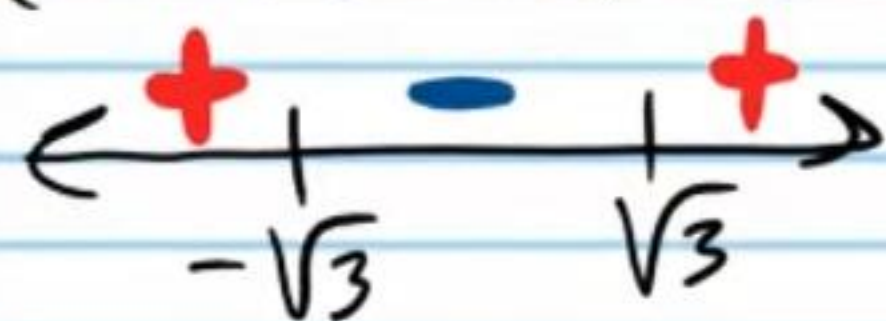
The Difference of Two Squares

• (23) $-3 + x^2 < 0$

$$x^2 - 3 < 0$$

$$x^2 - (\sqrt{3})^2 < 0$$

$$(x + \sqrt{3})(x - \sqrt{3}) < 0$$



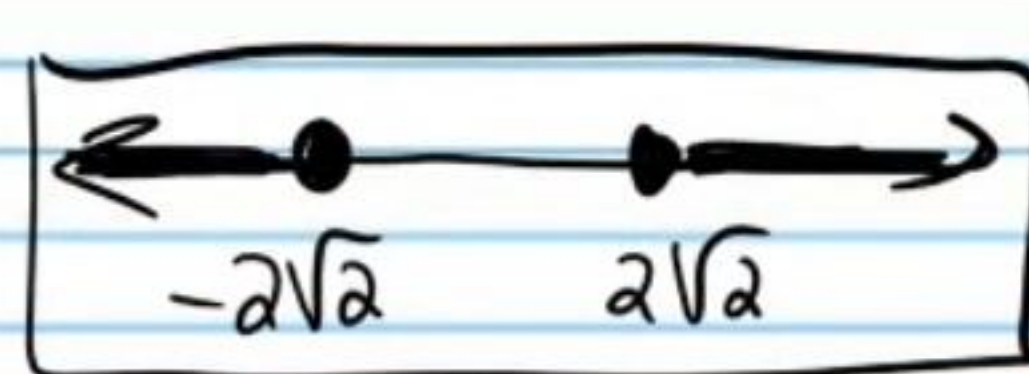
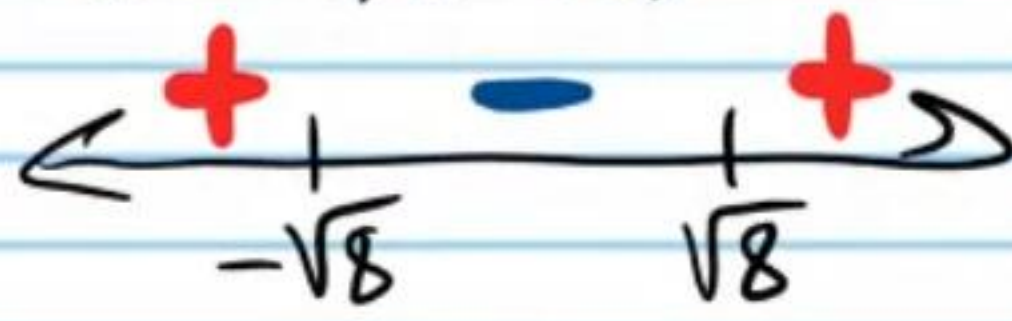
$$(-\sqrt{3}, \sqrt{3})$$

• (24) $x^2 \geq 8$

$$x^2 - 8 \geq 0$$

$$x^2 - (\sqrt{8})^2 \geq 0$$

$$(x + \sqrt{8})(x - \sqrt{8}) \geq 0$$



$$(-\infty, -2\sqrt{2}] \cup [2\sqrt{2}, \infty)$$

Using the Quadratic Formula

• (25) $x^2 + 5x \geq 1$

$$x^2 + 5x - 1 \geq 0$$

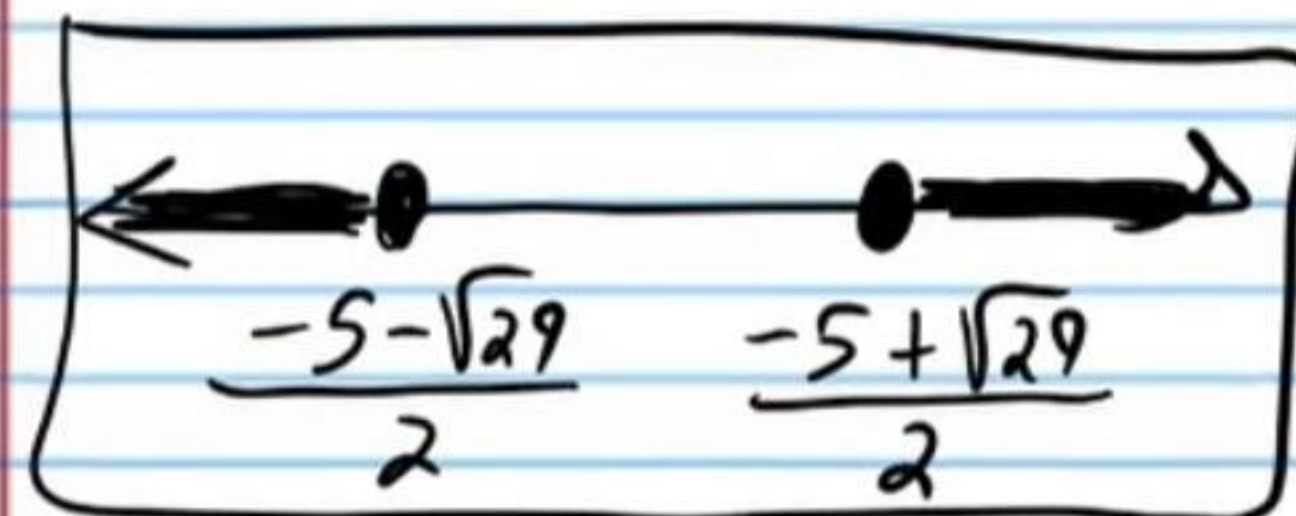
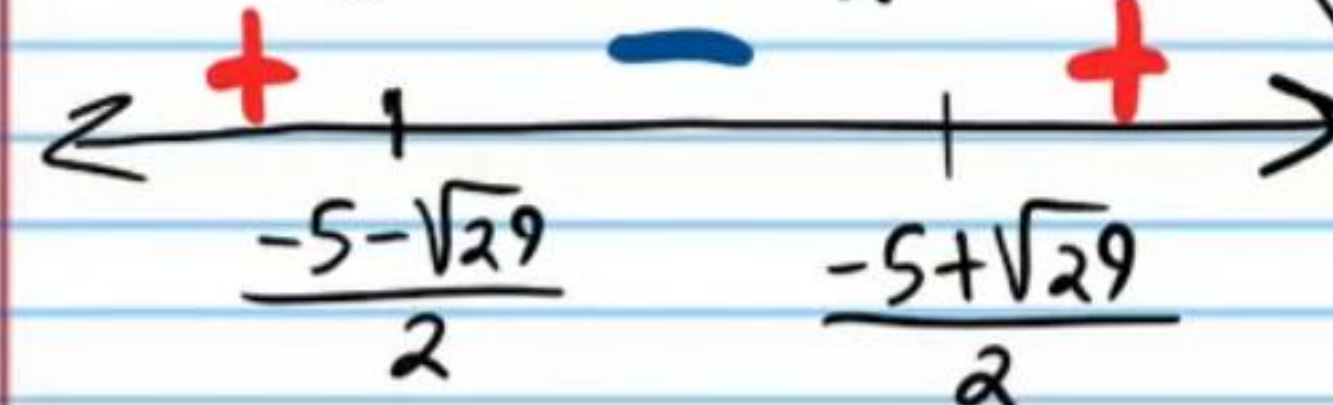
$$D = b^2 - 4ac = 5^2 - 4(1)(-1) = 29$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$x = \frac{-5 \pm \sqrt{5^2 - 4(1)(-1)}}{2(1)}$$

$$x = \frac{-5 \pm \sqrt{29}}{2}$$

$$(x - \frac{-5 + \sqrt{29}}{2})(x - \frac{-5 - \sqrt{29}}{2}) \geq 0$$



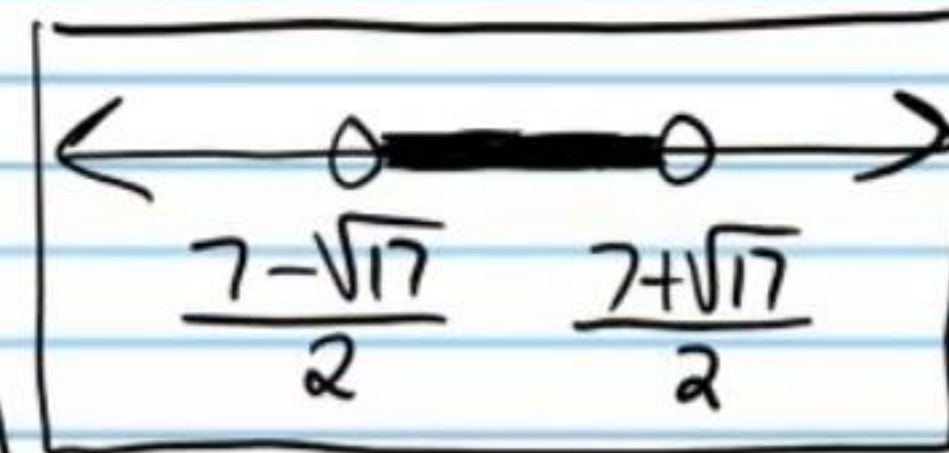
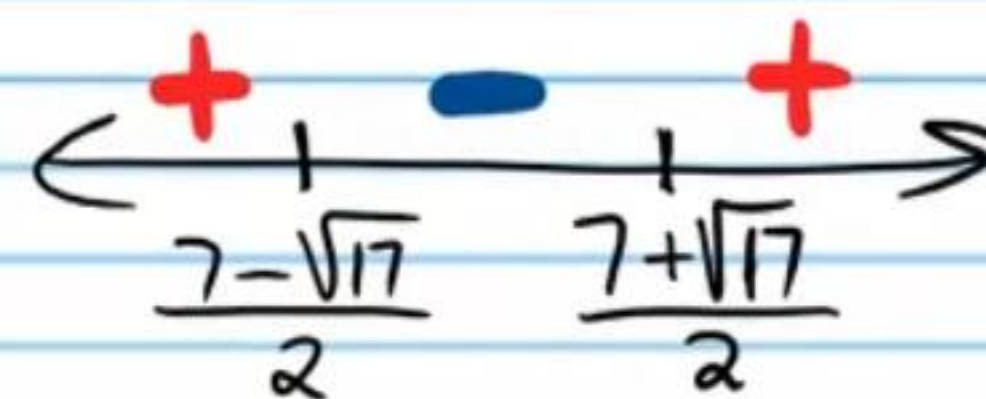
$$(-\infty, \frac{-5 - \sqrt{29}}{2}] \cup [\frac{-5 + \sqrt{29}}{2}, \infty)$$

• (26) $x^2 < 7x - 8$

$$x^2 - 7x + 8 < 0$$

$$x = \frac{-(-7) \pm \sqrt{(-7)^2 - 4(1)(8)}}{2(1)}$$

$$x = \frac{7 \pm \sqrt{17}}{2}$$



$$(\frac{7 - \sqrt{17}}{2}, \frac{7 + \sqrt{17}}{2})$$

Factoring the Sum or Difference of Two Cubes

Factoring the Sum or Difference of Two Cubes

$$a^3 + b^3 = (a+b)(a^2 - ab + b^2)$$

$$a^3 - b^3 = (a-b)(a^2 + ab + b^2)$$

• (27) $x^3 \geq -8$

$$x^3 + 8 \geq 0$$

$$p: 8 \Rightarrow 1, 2, 4, 8$$

$$q: 1 \Rightarrow 1$$

$$\frac{p}{q}: \pm 1, \pm 2, \pm 4, \pm 8$$

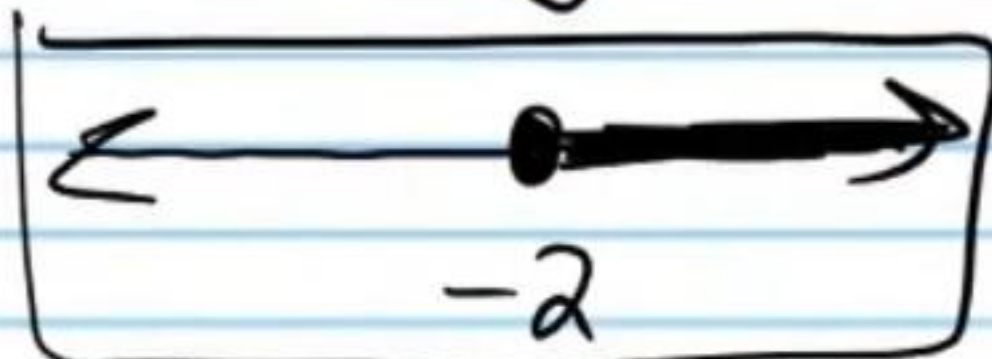
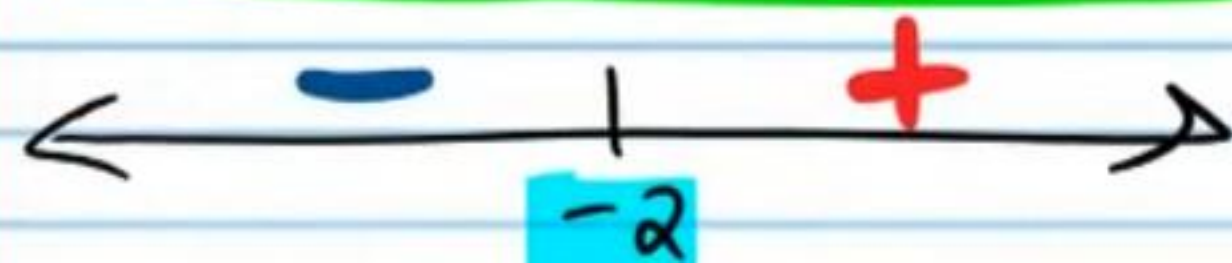
$$1 \Rightarrow \text{X}, -1 \Rightarrow \text{X}$$

$$2 \Rightarrow \text{X}, -2 \Rightarrow \checkmark$$

$$\begin{array}{r|rrrr} -2 & 1 & 0 & 0 & 8 \\ & & -2 & 4 & -8 \\ \hline & 1 & -2 & 4 & 0 \end{array}$$

$$(x+2)(x^2 - 2x + 4) \geq 0$$

$$D = b^2 - 4ac = (-2)^2 - 4(1)(4) = -12 \text{ X}$$



$$[-2, \infty)$$

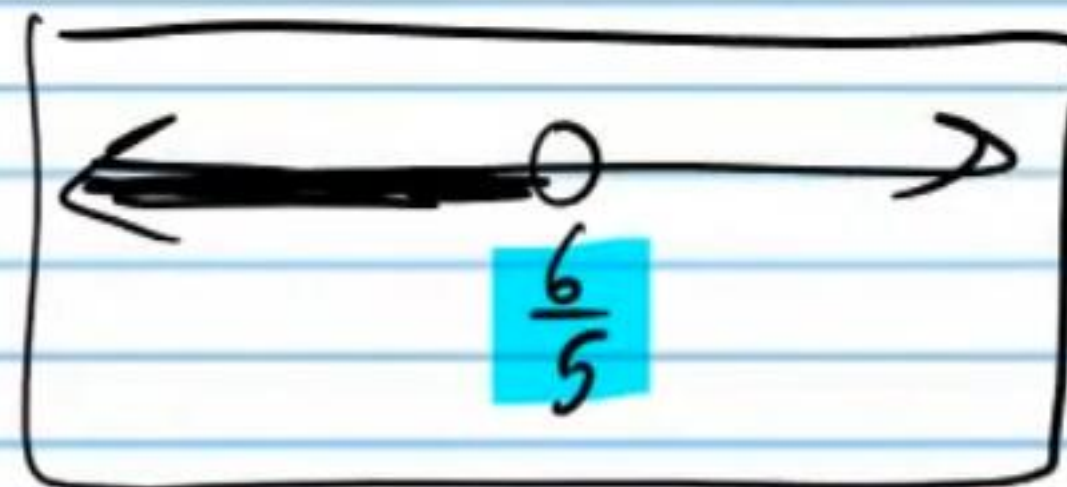
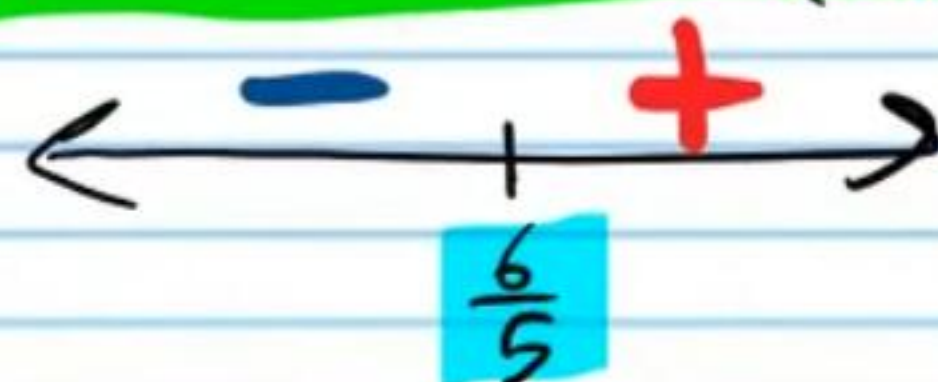
(28) $125x^3 - 216 < 0$

$$(5x)^3 - 6^3 < 0$$

$$(5x-6)((5x)^2 - (2x)(6) + 6^2) < 0$$

$$(5x-6)(25x^2 - 12x + 36) < 0$$

$$D = b^2 - 4ac = (-12)^2 - 4(25)(36) = -3,456 \text{ X}$$



$$(-\infty, \frac{6}{5})$$

• (29) $0 > x^3 - 125$

P: $125 \Rightarrow 1, 5, 25, 125$

q: $1 \Rightarrow 1$

$\frac{p}{q}: \pm 1, \pm 5, \pm 25, \pm 125$

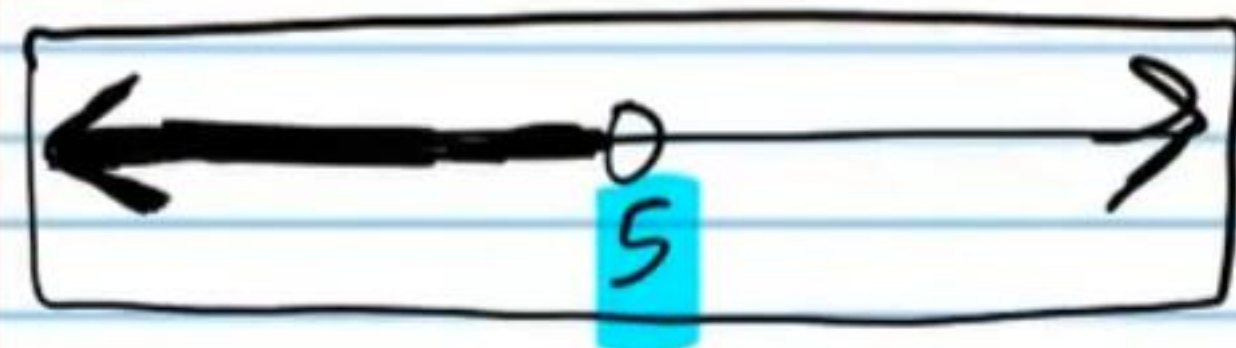
$1 \Rightarrow \text{X}$ $-1 \Rightarrow \text{X}$

$5 \Rightarrow \checkmark$

$$\begin{array}{r|rrrr} 5 & 1 & 0 & 0 & -125 \\ & & 5 & 25 & 125 \\ \hline & 1 & 5 & 25 & 0 \end{array}$$

$0 > (x-5)(x^2+5x+25)$

$D = b^2 - 4ac = 5^2 - 4(1)(25) = 25 - 100 = -75 \text{ X}$



$(-\infty, 5)$

(30) $8x^3 \leq -1$

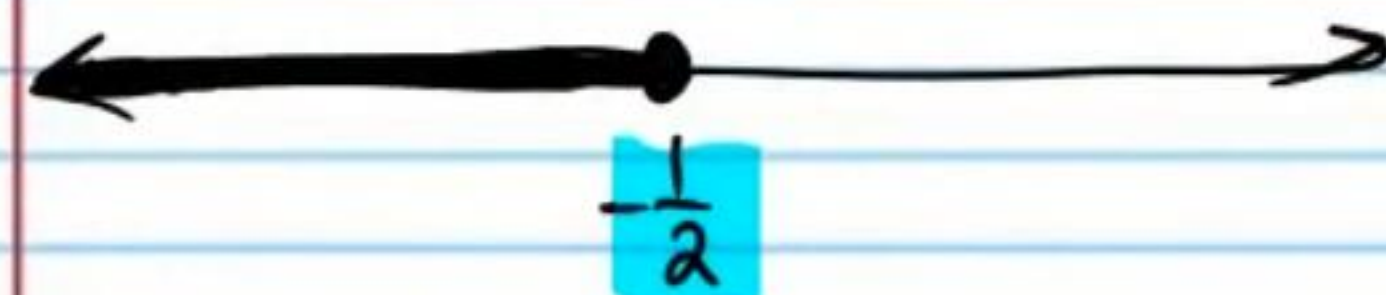
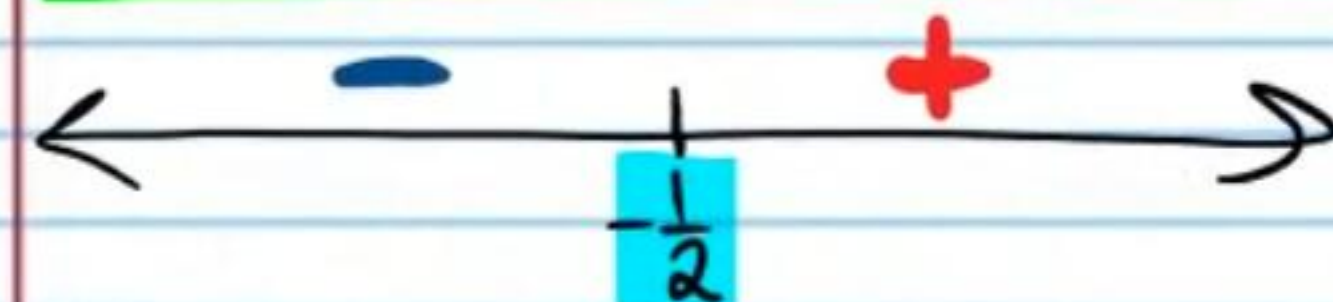
$8x^3 + 1 \leq 0$

$(2x)^3 + 1^3 \leq 0$

$(2x+1)((2x)^2 - 2x(1) + 1^2) \leq 0$

$(2x+1)(4x^2 - 2x + 1) \leq 0$

$D = b^2 - 4ac = (-2)^2 - 4(4)(1) = 4 - 16 = -12 \text{ X}$



$(-\infty, -\frac{1}{2}]$